

Digital I/O field Interface Communication Protocol

(AT520/AT712/AT705-IO/ AT812)

(V 1.0)

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by

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ATOP's AT520, AT712, AT705-IO and AT812 all have Digital Input(DI) and Digital Output(DO) design. All of them share the same communication protocol, Only the difference is the number of DI/DO. AT520 has 4 DIs and 4 Dos, the other AT712 , AT705-IO and AT812 all have one DI and one DO design.

1. COMMAND SET

1.1 Product ID of I/O Devices

I/O TYPE : 0x30 → DI
 0x31 → DO
 0x34 → DI_AUTO response mode
 0x35 → DO_BLINK mode

1.2 Reading I/O Status :

※ Application Program(AP) sends reading command to TCP/IP controller :

CCB LEN(L) = 08H	
CCB LEN(H) = 00H	
MESSAGE TYPE = 60H	Port 1, 61H :port 2
Reserved	
Reserved	
Reserved	
SUB-COMMAND = 3CH	
SUB-NODE = {01H ~ C8H}	Max. device address (up to 200)

※ TCP/IP controller reports the current Digit I/O status when it receive Sub-command = 3CH:

CCB LEN(L) = 0EH	
CCB LEN(H) = 00H	
MESSAGE TYPE = 60H	Port 1, 61H :port 2
Reserved	
Reserved	
Reserved	
SUB-COMMAND = 3CH	
SUB-NODE	0x01~0xC8
Reserved	00H
DI/DO type	
Data #1	ASCII of '0'~'9','A'~'F'
:	ASCII of '0'~'9','A'~'F'
Data #4	ASCII of '0'~'9','A'~'F'

DI/DO Type :

DI : 30H
 DO : 31H
 DI_AUTO : 34H

Data #1 : DI/DO Channel 5th to 8th
 Data #2 : DI/DO Channel 1st to 4th
 Data #3 : DI/DO Channel 13th to 16th
 Data #4 : DI/DO Channel 9th to 12th

DI type :

Data #1 = high nibble of low byte of DI status
 Data #2 = low nibble of low byte of DI status

Data #3 = high nibble of high byte of DI status
Data #4 = low nibble of high byte of DI status

Example: ch16ch9.....ch1
0000 1100 1001 1111 ---> 1: means ON, 0: means OFF
ASCII '0' 'C' '9' 'F'
means -----> ch 1, 2, 3, 4, 5, 8, 11, 12 is on ch 6, 7, 9, 10, 13, 14, 15, 16 is off

data #1 = 0x39('9')
data #2 = 0x46('F')
data #3 = 0x30('0')
data #4 = 0x43('C')

DO type :

Data #1 = high nibble of low byte of DO status
Data #2 = low nibble of low byte of DO status
Data #3 = high nibble of high byte of DO status
Data #4 = low nibble of high byte of DO status

Example: ch16ch9.....ch1
0011 0100 0001 0010 --> 1: means ON, 0: means OFF
ASCII '3' '4' '1' '2'
means -----> ch 2, 5, 11, 13, 14 is on ch 1, 3, 4, 6, 7, 8, 9, 10, 12, 15, 16 is off

data #1 = 0x31('1')
data #2 = 0x32('2')
data #3 = 0x33('3')
data #4 = 0x34('4')

※ TCP/IP controller reports the current Digit I/O status for DI_AUTO:

CCB LEN(L) = 12H	Port 1
CCB LEN(H) = 00H	
MESSAGE TYPE = 60H	
Reserved	
Reserved	0x01~0xC8
Reserved	
SUB-COMMAND = 3CH	
SUB-NODE	
Reserved	34H:DI-AUTO
I/O type = 34H	
Data #1 (Now Status)	
:	
Data #4 (Now Status)	
Data #5 (last time Status)	
:	
Data #8 (last time Status)	

DI-AUTO type:

Data #1 ~ Data #4 = the current status for channel 1 to 16
Data #5 ~ Data #8 = last time status for channel 1 to 16

1.3 Enable the automatic reporting DI status: (DI_AUTO)

CCB LEN(L) = 0x0D	Port 1
CCB LEN(H) = 0x00	
MESSAGE TYPE = 60H	
Reserved	

Reserved	
Reserved	
SUB-COMMAND = 3AH	CONF CMD
SUB-NODE	0x01~0xC8/ 0xFC
40H	
1BH	
1BH	
11H	
'A'	Enable auto mode

1.4 Disable the automatic reporting DI status (DI_AUTO)

CCB LEN(L) = 0x0D	
CCB LEN(H) = 0x00	
MESSAGE TYPE = 60H	Port 1
Reserved	
Reserved	
Reserved	
SUB-COMMAND = 3AH	CONF CMD
SUB-NODE	0x01~0xC8/ 0xFC
40H	
1BH	
1BH	
11H	
'N'	Disable auto mode

1.5 Write DO Value with mask

CCB	
CCB LEN(L) = 0x11	
CCB LEN(H) = 0x00	
MESSAGE TYPE = 60H	Port 1
Reserved	
Reserved	
Reserved	
SUB-COMMAND = 3DH	
SUB-NODE	0x01~0xC8 or 0xFC
Reserved	
Low byte data #1	High nibble of mask value
data #2	Low nibble of mask value
data #3	High nibble of DO value
data #4	Low nibble of DO value
High byte data #5	High nibble of mask value
data #6	Low nibble of mask value
data #7	High nibble of DO value
data #8	Low nibble of DO value

Note: The DO bit value is changed when the corresponding mask bit is '0'.

Processing method:

A= original DO value

B= DO value specified by AP

M= Mask value specified by AP

Result = (A and M) or (B and (not M))

Example 1: ch 1 ~ ch 8 are ON

Data #1	0x30('0')	ch 5 ~ ch 8 enabled (no mask)
Data #2	0x30('0')	ch 1 ~ ch 4 enabled (no mask)
Data #3	0x46('F'),	ch 5 ~ ch 8 is all ON
Data #4	0x46('F'),	ch 1 ~ ch 4 is all ON
Data #5	0x46('F'),	ch 9 ~ ch 12 disabled(mask is ON)
Data #6	0x46('F'),	ch 13 ~ ch 16 disabled (mask is ON)
Data #7	Any value	
Data #8	Any value	

Example 2: ch 1 is ON only

Data #1	0x46('F')	ch 5 ~ ch 8 disabled(mask is ON)
Data #2	0x30('0')	ch 1 ~ ch 4 enabled (no mask)
Data #3	Any value	
Data #4	0x31('1'),	ch 1 is ON, ch2, ch3, ch4 are OFF.
Data #5	0x46('F'),	ch 9 ~ ch 12 disabled(mask is ON)
Data #6	0x46('F'),	ch 13 ~ ch 16 disabled (mask is ON)
Data #7	Any value	
Data #8	Any value	

Example 3: ch 4 is ON only

Data #1	0x46('F')	ch 5 ~ ch 8 disabled(mask is ON)
Data #2	0x30('0')	ch 1 ~ ch 4 enabled (no mask)
Data #3	Any value	
Data #4	0x38('8'),	ch 4 is ON, ch1, ch2, ch3 are OFF.
Data #5	0x46('F'),	ch 9 ~ ch 12 disabled(mask is ON)
Data #6	0x46('F'),	ch 13 ~ ch 16 disabled (mask is ON)
Data #7	Any value	
Data #8	Any value	

Example 4: ch 3 & ch 4 are ON

Data #1	0x46('F')	ch 5 ~ ch 8 disabled(mask is ON)
Data #2	0x30('0')	ch 1 ~ ch 4 enabled (no mask)
Data #3	Any value	
Data #4	0x43('C'),	ch 3 & ch4 are ON, ch1, ch2 are OFF.
Data #5	0x46('F'),	ch 9 ~ ch 12 disabled(mask is ON)
Data #6	0x46('F'),	ch 13 ~ ch 16 disabled (mask is ON)
Data #7	Any value	
Data #8	Any value	

Example 5: ch 2 & ch 4 are ON

Data #1	0x46('F')	ch 5 ~ ch 8 disabled(mask is ON)
Data #2	0x30('0')	ch 1 ~ ch 4 enabled (no mask)
Data #3	Any value	
Data #4	0x41('A'),	ch 2 & ch4 are ON, ch1, ch3 are OFF.
Data #5	0x46('F'),	ch 9 ~ ch 12 disabled(mask is ON)
Data #6	0x46('F'),	ch 13 ~ ch 16 disabled (mask is ON)
Data #7	Any value	
Data #8	Any value	

1.6 Reading DO_Blink Status

- ※ Application Program(AP) sends reading command to TCP/IP controller :

CCB	
CCB LEN(L) = 0x0A	
CCB LEN(H) = 0x00	
MESSAGE TYPE = 0x60	
Reserved	E
Reserved	
Reserved	
SUB-COMMAND = 0x3F	DO BLINKING CMD
SUB-NODE	0x01~0xC8 or 0xFC
Reserved	
CMD = 'A'	READ DO BLINKING CMD

- ※ TCP/IP controller reports the current Digit I/O status when it receive Sub-command = 3FH:

CCB	
CCB LEN(L) = 0x11	
CCB LEN(H) = 0x00	
MESSAGE TYPE	0x60
Reserved	DCA NODE
Reserved	DTA NODE
Reserved	
SUB-COMMAND = 0x3F	RESPONSE CMD
SUB-NODE	0x01~0xC8 or 0xFC
Reserved	
'A'	
0x35	0x35:DO BLINKING
Data #1	ASCII '0'~'9','A'~'F'
:	ASCII '0'~'9','A'~'F'
Data #4	ASCII '0'~'9','A'~'F'
Data #5 = low byte timer	Default : 0x80
Data #6 = high byte timer	Default : 0x00

The detail data format of each I/O type is shown as

follow :

DO_BLINK type :

data #1 = high nibble of low byte of DO status
data #2 = low nibble of low byte of DO status
data #3 = high nibble of high byte of DO status
data #4 = low nibble of high byte of DO status

data #5 = low byte of the blinking timer.

data #6 = high byte of the blinking timer.

Example: ch16 .ch9 ..ch1

0011 0100 0001 0010

1: means blinking ,

0: means no blinking

data #1 = 0x31('1')

data #2 = 0x32('2')

data #3 = 0x33('3')

data #4 = 0x34('4')

data #5 = 0x80

data #6 = 0x00

(0x0080 x 1msec = 128msec)

1.7 Write DO_BLINK Value with mask

CCB LEN(L) = 0x12	
CCB LEN(H) = 0x00	
MESSAGE TYPE = 0x60	Port 1
Reserved	
Reserved	
Reserved	
SUB-COMMAND = 0x3F	DO BLINKING CMD
SUB-NODE	0x01~0xC8 or 0xFC
Reserved	
CMD = 'B'	WRITE DO BLINKING CMD
Low byte data #1	High nibble of mask value
data #2	Low nibble of mask value
data #3	High nibble of DO value
data #4	Low nibble of DO value
High byte data #5	High nibble of mask value
data #6	Low nibble of mask value
Data #7	High nibble of DO value
Data #8	Low nibble of DO value

Note: The DO_BLINK bit value is changed when the corresponding mask bit is '0'.

Processing method:

A= original DO_BLINK value
 B= DO_BLINK value specified by AP
 M= Mask value specified by AP
 Result = (A and M) or (B and (not M))

Example:

data #1 = 0x30('0'), data #2 = 0x30('0')
 data #3 = 0x46('F'), data #4 = 0x46('F')
 result = 0xff -> ch1 ~ch8 all 'ON'

data #5 = 0x46('F'), data #6 = 0x46('F')
 data #7 = Any value, data #8 = Any value
 result = ch9 ~ ch16 no change

1.8 Setup DO_BLINK Blinking timer

CCB	
CCB LEN(L) = 0x0A	
CCB LEN(H) = 0x00	
MESSAGE TYPE = 0x60	Port 1
Reserved	
Reserved	
Reserved	
SUB-COMMAND = 0x3F	DO BLINKING CMD
SUB-NODE	0x01~0Xfa
Reserved	
CMD = 'C'	SETUP DO BLINKING TIMER
Timer low byte	Blinking timer
Timer high byte	

Blinking timer = high byte x 256 + low byte

Default : Timer low byte = 0x80.

Timer high byte = 0x00.

2. AT705-IO

The default mode for AT705-IO is its DI and DO are working independently from its LED display and confirmation button. However, it also could be configured the linkage between DI/DO and illuminated button to work Synchronously.

CCB LEN (L) = 0AH	Data length of this block
CCB LEN (H) = 00H	
MESSAGE TYPE = 60H	60H : port 1
Reserved	
Reserved	
Reserved	
SUB-COMMAND = 1F H	
SUB-NODE = {01H ~ FAH}	0x01~0xFA (address) or 0xFC (Broadcasting)
DATA[0] = 11H	
DATA[1] = Configuration	

DATA[1] =

Bit0: default reserved-1	1-enable/0-disable
Bit1: default reserved-2	1-enable/0-disable
Bit2: default reserved-3	1-enable/0-disable
Bit3: DIO Linkage	1-enable/0-disable
Bit4: no used	
Bit5: no used	
Bit6: no used	
Bit7: no used	

Note (1) : Bit 3 : DIO Linkage

This bit can enable or disable the linkage between AT705-IO's DI/DO and illuminated button. When it is enabled, then the trigger coming from DI (such as sensor) will cause the illuminated button to be confirmed instead of pushing the confirmation button. Or when turn on the illuminated button also will have signal output to its DO to trigger the connected device(such as turn on external LED light).

DATA[2] = 55H -----> Store the color configuration into EEPROM



Warning : EEPROM has the life cycle for writing. Thus we don't recommend to save every new configuration into EEPROM each time.

Please set 100 ms delay time to allow configuration value to write into EEPROM.